Conjecture 00000002854: full generic Jacobian rank does not give global SEM identifiability

GitHub: gaochengzhecpu

Abstract

A one-factor Gaussian structural equation model has observed variance λ^2+1. Its parameter Jacobian is nonsingular away from zero, but the distinct parameters λ and $-\lambda$ give identical observed covariance matrices and identical complete observed laws. Thus full generic Jacobian rank does not imply strict global identifiability, contrary to the proposed equivalence.

The model and parameter convention

Let H and E be independent standard real Gaussian variables, and consider the structural equations

$$H = U_H, \qquad Y = \lambda H + U_Y, \qquad (U_H, U_Y) = (H, E).$$

Only Y is observed. The sole unknown parameter is the labelled loading $\lambda \in \mathbb{R}$. Both exogenous means are fixed at zero; both variances are fixed at one; the exogenous covariance is fixed at zero. Thus this is a one-edge latent-variable Gaussian SEM with a fixed error variance, rather than an example with an unnormalized latent scale or an additional unobserved variance parameter.

We use strict global identifiability of the actual real parameter: equality of observed laws must force equality of parameter values. In the algebraic covariance formulation, this requires injectivity of the observed covariance parametrization. We do not identify positive and negative loadings, nor impose a positive-loading convention. Taking a quotient by latent sign or restricting to $\lambda > 0$ changes the parameter problem and removes the ambiguity below. Such alternatives do not establish the unqualified global equivalence stated in the source.

Theorem 1. *The model has full generic Jacobian rank but is not strictly globally identifiable. This failure occurs at every nonzero real parameter.*

Proof. The loading row and exogenous covariance matrix are

$$L(\lambda) = \begin{pmatrix} \lambda & 1 \end{pmatrix}, \qquad \Sigma = I_2.$$

The entire observed covariance matrix is therefore

$$C(\lambda) = L(\lambda)\Sigma L(\lambda)^{\mathsf{T}} = \begin{pmatrix} \lambda^2 + 1 \end{pmatrix}.$$

The parameter space and covariance coordinate space are both one-dimensional. The genuine Jacobian is $J(\lambda) = (2\lambda)$, which has rank one for every $\lambda \neq 0$. Its exceptional set is the proper algebraic set $\{0\}$, so the rank deficit at the generic point is zero.

On the other hand, $C(\lambda) = C(-\lambda)$ for every λ. For $\lambda \neq 0$ these are distinct parameter values. For example, 1 and -1 both give observed covariance (2). The ambiguity also persists in the complete observed law: (H, E) and $(-H, E)$ have the same product Gaussian distribution, and

$$(-\lambda)(-H) + E = \lambda H + E.$$

Hence Y has exactly the same distribution at λ and $-\lambda$. This is a genuine global ambiguity, including at the generic nonsingular parameters, and not a failure confined to zero. □

Relation to the source

Both source languages propose global identifiability if and only if the generic Jacobian rank deficit is zero. The single-factor family above refutes its sufficiency direction. The remaining source assertion about compensating rank deficits with instrumental-variable moments cannot repair this equivalence as written: in this example there is already no generic rank deficit to compensate. No assertion is made about a specified instrumental-variable model.

The distinction is the familiar one between local information and global uniqueness. On either half-line $\lambda > 0$ or $\lambda < 0$, the variance map is injective. On the stated parameter space $\mathbb{R}$, it is generically two-to-one. Sign equivalence would give a different identifiability question, which this disproof explicitly does not address.

Formal verification and scope

The Lean file defines the actual rectangular loading matrix, the fixed identity exogenous covariance matrix, and matrix multiplication $L\Sigma L^{\mathsf{T}}$. It computes every entry of the resulting observed covariance matrix. For the affine-loading SEM family $Y = (a\lambda + b)H + E$, the parameter map is its sole covariance entry, and the Jacobian is defined using the actual real derivative `deriv`. Lean proves the derivative and nonsingular determinant for the counterexample $a = 1$, $b = 0$, with exceptional set $\{0\}$.

The proposition `RankCriterion` universally asserts the source's equivalence on this actual SEM subclass; `GloballyIdentifiable` means injectivity of the entire observed covariance matrix map. The final theorem is

`Conjecture2854.disprove_rank_criterion : ¬RankCriterion.`

Lean separately constructs the real product measure $N(0,1) \otimes N(0,1)$ using Mathlib's Gaussian distribution and proves equality of its pushforwards under the two observed-output maps. Thus the complete-law sign ambiguity is also formalized, rather than inferred only from equal numerical variances. The covariance formula uses the standard fixed exogenous covariance specification of the SEM; the file does not separately evaluate Gaussian moment integrals. It does formally establish all matrix, derivative, generic nonsingularity, noninjectivity, and observed-law identities used in the argument. The audited main results use only `propext`, `Classical.choice`, and `Quot.sound`.

Source and reproduction

The unchanged bilingual source is included as `SOURCE.md`, from upstream commit

`b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

The public statement is conjecture 00000002854. Use Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From the `lean` directory, run `lake build`, followed by

`lake env lean -DwarningAsError=true Main.lean`.

This draft was developed by a delegated AI agent with a self-review; the coordinating agent reviews it separately.